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CORPORATE COMPUTER CRIME MANAGEMENT: A REASEARCH PERSPECTIVE H8

Corporate Computer Crime Management: a research perspective

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ABSTRACT

This paper analyses the problems posed by the emergence of computer crime and the possible avenues for its control and management. It analyses the various approaches to the study of crime. Based on these it proposes a research perspective to manage corporate computer crime.

INTRODUCTION

There is widespread agreement that computer crime is a growing problem for international society today. We are constantly reminded of spectacular intrusions into supposedly secure computer installations for a variety of illicit purposes, including theft, fraud and sabotage. Such is the concern that the attentions of international organisations such as the Organization for Economic and Co-operation and Development (OECD, 1986), the International Chamber of Commerce (ICC, 1988) and the Council of Europe (Council of Europe, 1990), amongst many others, have focused on the question.

Evidence coming from across the globe indicates that computer related crime afflicts practically all nations, even those we might consider unlikely - recently from the People's Republic of China news has come of the execution of a hacker who embezzled from the Agricultural Bank of China a sum worth £122,000 (Computing, 1993). While it is clear that considerable sums of money are being illegally obtained by computer-related crime, it is difficult to find reliable figures and extreme caution is needed. Two countervailing tendencies serve to confound the picture. On the one hand figures may be exaggerated by interests such as auditors and security consultants, undoubtedly to highlight the need for their services. On the other hand there is the desire by the victims of computer-related crime to cover up the incident and avoid

adverse publicity. This is especially so in the vulnerable finance and banking sectors of the international economy (Essinger, 1992). Figures coming from the USA suggest anything up to yearly damages of \$2,000 million could be involved, although something in the region of \$145-730 million seems to be more realistic. In the United Kingdom, even six years ago, the Local Government Audit Commission found that one in ten of the 1200 organisations surveyed admitted to suffering from computer fraud, with an average value of £47,000 loss for each case. A conservative estimate of the loss in computer frauds for the United Kingdom has been given as £30 million, excluding damages caused by lost or disclosed data.

Against this background of computer-related crime, lawyers and others are still coping with the new problems raised by the ubiquitous reliance on computer technology by businesses and organisations. Data held in computer systems is extremely valuable to its owners, so much so that increasingly it is common to claim exclusive rights in relation to third parties. However it appears that far less security is applied to data held in computer systems than is the case for data held in manual systems. Office workers are familiar with the security requirements of a filing cabinet but not necessarily those with a computer system. Telecommunications and information networks have overcome the distance factor and so the store of data in one computer system can be attacked from outside national borders.

UNDERSTANDING COMPUTER CRIME

The importance of understanding the nature of crime cannot be under-estimated. The development of computer technology and society's growing reliance upon it to transact business and store valuable information has resulted in the creation of new forms of illicit behaviour. Hence the understanding of computer crime as a form of white collar crime gains significance. If unchecked these developments could seriously affect the functioning of computer-using organisations. Controlling and managing computer crime is a task facing not only the corporate world but also the national and international community.

Research on the nature of crime has suggested three categories. First, a situation where crime originates due to *personal factors*. Second, when a criminal act is performed because of the *work situation*. Third, when the corporate environment offers *opportunities* for such an act. Researchers have proposed various theories of crime which either fall into one of the above categories or encompass all of them. Sutherland (1949), for example, proposed the theory of differential association. He takes the "...point of view of the process by which a person is initiated into crime". The approach not only considers the personal factors of an individual but also looks at the work environment.

Researchers have adopted different perspectives in analyzing crime arising due to personal factors. Croall (1992) regards greed as a basis for crime. He attributes blame on the selfishness and individualism inherent in the values of a capitalist society.

Cressey (1986) on the other hand takes a more individualistic approach. According to him embezzlement occurs because of personal, non-sharable financial problems. On basis of the interviews with retired middle management executives (Clinard, 1983), it becomes obvious that there is a direct co-relation between the personal factors of top management executives and incidents of crime. Clinard's analysis revealed that the top management executives' personality and character often influences the internal structure of the corporation. This viewpoint is further supported by the notion of a *system of ideology* that operates at the top management level (Mintzberg, 1983). The ideology of the top management creates an informal norm structure in the organization. Any influence, internal or external, on this norm structure results in the individuals being alienated. In such a situation organisations that rely heavily on information technology, become vulnerable.

However personal factors provide inadequate explanations about computer crimes (see Boisjoly *et al* 1992; Passas, 1990; Braithwaite, 1985; Schraeger and Short, 1977). Hence it is important to provide a macro-sociological rather than an individualistic explanation. In providing such an explanation, risks associated with social and cultural systems of an organization are to be identified. This socio-organisational environment reflects on the structure of the organization. Such risks increase with the complexity and the geographical spread of the corporation. Typically subsidiaries of the organization can be used by the top management to circumvent control at the operational level (see Croall, 1992; Clarke, 1990; Braithwaite, 1984; 1985; Aubert, 1977). Since most fraud depends on collusion or compliance of management and staff, it becomes evident that the quality of management is the most important environmental factor. In a study conducted by Clinard and Yeager (1980), it was found that nearly 40% of the large corporations had no record of any offence. Therefore they regard the culture of an organization as playing an important role in the performance of a criminal act. These cultures are formed on basis of the attitude of the top management and the technological and social organization of work (Croall, 1992; Mintzberg, 1983).

Crimes are often committed because of the opportunities that an environment may offer. Scraton and South (1984) recognise social class and occupational position as different types of opportunities for committing offences. They associate crimes of operational workers with low pay and oppressive working conditions. They also suggest that operational level workers are often subjected to a high level of surveillance and that their offences are less tolerated than those who are high up in the management hierarchy. Shapiro (1990) attributes reasons for criminal acts as the differential distribution of trust. Employees who are more likely to be trusted have a greater opportunity to abuse that trust.

With this background any criticism of the existing approaches is subject to severe counter-criticism. However in order to control and manage computer crime in corporations, it is essential to view organisations from an alternative perspective. Anthony (1986) writes, "the authority of management must rest upon a moral base, secure in a concern for the integrity and the good of the community that it governs".

He views organisations as moral communities held together by informal relationships. His argument is based on the concept that:

the foundation of managerial authority, its legitimation by those subordinate to it cannot be assured by other means than the acceptance by management of its responsibility to the general community and the government of its own.

This gives insight into the manner in which organisations should be governed. Indeed Anthony has a point in viewing organisations as moral communities with the prevalence of informal structures. Following Anthony's approach it will be of value to identify where responsibility resides and the manner in which it is attributed.

COMPUTER CRIME MANAGEMENT

Legislative control

Control can be implemented at two levels. At a macro level by enacting relevant laws and at a micro level by adopting better management practices. For the international community the pressing question is how to provide legislative powers to control this new affliction. The response from national jurisdictions has ranged from either adapting existing legislation or developing and supplementing it. In Europe, countries such as Austria, Denmark, France, Germany, Greece, Liechtenstein, Norway and Sweden had made extensive amendments to their substantive criminal law by 1990, while since then Spain, Portugal and the United Kingdom have introduced legislation. Outside the Council of Europe Australia, Canada, Japan, Switzerland, and the USA (at state and federal level) all have introduced new statutes. Italy remains somewhat behind the pack in introducing computer crime legislation.

In view of the growing incidence of transborder crime and the need to collaborate, the Council of Europe's European Committee on Crime Problems proposed in 1990 a three step approach towards the international control of computer crime:

- stipulating what acts constitute offences by amendments and supplements to substantive criminal law;
- effective prosecution, *inter alia*, by possibly adapting domestic criminal procedural law and related provisions;
- improving international collaboration.

With regard to the first of these proposals, they produced a set of guidelines for legislation on computer crime and these were seen as the first steps in moving towards a European criminal policy in this area. They declined to define the key terms "computer crime" and "computer-related crime". Instead they offered a functional definition, as did a previous OECD report, in the form of two lists, a minimum list and optional list for offences necessary for a uniform criminal policy on legislation concerning computer-related crime.

The offences in this minimum list furnish the basis for crimes which should be specified in local legislation. If such a concept were to be acceptable it would encompass the aim of the first step in the European Committee's approach to criminal policy in this area, effectively dealing with the common approach on the substantive questions. In addition to these we could include also theft of the computer itself, theft of data, software and semi-conductor chips, and theft of services provided by the system (London, 1992). There seems to be a growing realization that existing law cannot easily be applied to cover the area of computer crime, and that additional substantive legislation is required.

Better management practices

Better management practice is an equally important issue in computer crime management. In order to develop such practices, it is essential to evaluate computer crime within the context of organisations. Ermann and Lundman (1992) view organisations as made up of a collection of people. These people occupy certain positions/roles. These roles are generally fixed, whereas people may change over a period of time. The roles have certain obligations and expectations of the people who occupy them. Thus organisations can be viewed in terms of their stable underlying patterns of behaviour (see Backhouse and Dhillon, 1993). Based on this viewpoint of an organization, the following deductions can be made:

- organisations can be viewed as communication systems (Dhillon and Backhouse, 1993; Dietz, 1992; McCall and Cousins, 1990; Liebenau and Backhouse, 1990). Any disruption in the communication process leads to security and integrity problems in an organization (Backhouse and Dhillon, 1993);
- a crime is committed because the role provides the occupier an opportunity to commit an offence;
- since a crime has been committed, the occupier of the role has abandoned the responsibilities, obligations and expectations of that role.

In viewing organisations as communication systems, we can place the essentially technical problems into their social and organizational context. Research into security is often concerned with technical issues such as systems architecture, hardware performance, databases and software design. This alternative research perspective allows us to integrate these issues with considerations of norms, purpose, and interpretation of information as regards the security of organisations.

Since a crime results in abusing the responsibilities and obligations of a role, the endeavour should be to elicit the *structures of responsibility*. One way of modeling these structures is the semantic schema, that represents organisations as ordered patterns of behaviour (see figure 1). Providing security is best seen as part of the management of such patterns of behaviour. In preparing a schema, it is important to identify the agents who determine what takes place, and what behavior is realized. Every agent, within the sphere of discourse has a determinate range of conduct possible. This range aggregates to the behaviours that are afforded by an environment. Taking an example from the domain of secure computing (Figure 1), the manner in

which the schema is prepared can be demonstrated. The first step is to identify the agents, those who can take responsibility for their actions and would generally be associated with purposeful behavior, for example providing access to a PC. Various roles within complex agents are assumed. Complex social agents, in fact, are not really a collection of parts, but are 'collectives' (Downes, 1984).

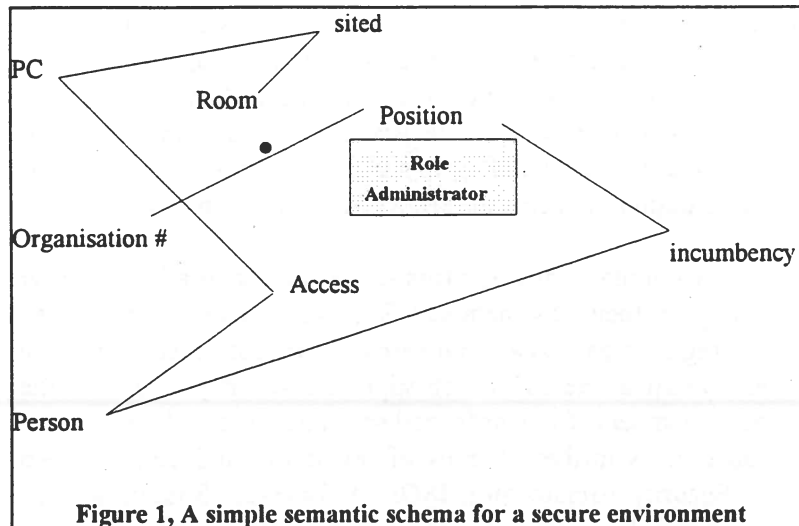


Figure 1 indicates the patterns of behavior that might be associated with the general area of data security. Notice that the representation does not refer to any specific rules or procedures employed by an organization but rather sketches the generic affordances that constrain any agent in this domain. Every organization that addresses the question of security of data will be engaged in realizing the patterns shown. A providential company will have careful policies designed to deal actively with these matters in a way that contributes to the overall confidentiality, integrity and availability of equipment, programs and data. Others will be less formal undertakings where these matters are not driven by rules and policies, but where instead decisions are taken by individuals in an informal manner. There may be norms governing the domain; for example a norm that gives access to a PC once a person has become the incumbent of a specified role. Norms could be strong or weak. The above case is an example of a strong norm. In a context where the norms are strong, the conduct of an agent will be constrained informally yet effectively. In such situations the norms may coincide with any rules that apply.

Thus in identifying the responsible agents and capturing the norms associated with each action, we are in a position to understand the underlying repertoires of behavior. By looking at the informal environment, the semantic schema is able to capture the structures in their cultural context. This enables the analyst to understand the object system better. In managing security in an organization, such an approach can aid in illuminating concepts such as attribution of blame, responsibility, accountability and authority.

Adequate security and reporting of breaches

One of the cornerstones of control and management of computer crime is the security policy that organisations have in place to protect their systems. These policies are designed to prevent, detect, withstand or react to attacks on systems. Such measures include technical controls (access controls, passwords) and organizational policies, including those for personnel (e.g. vetting). A key principle in assessing the resources to allocate to security is that the amount spent on security should be in proportion to the criticality of the system, cost of remedy and the likelihood of the breach of security occurring. It is necessary for the management of organisations to adopt appropriate security policies to protect themselves from claims of negligent duty and also to comply with the requirements of data protection legislation. The semantic schema, introduced above, not only helps in understanding the semantics of key terms in the legislation but also in the construction of norms for security management policies.

Professional bodies including accountants and auditors are beginning to add computer security to the required training for their new members. Security requirements can also be specified by bodies which regulate particular commercial or industrial sectors, such as the Securities Investment Board in the UK which stipulates security levels for the information systems of their members. Standards bodies, such as the International Standards Organization, function as further sources of regulation and control. An example is the standard on Security Architecture, ISO/DIS 7498-2. Situations arise where organisations that may exchange data will only undertake to do so when both parties can demonstrate compliance with an appropriately specified security standard. Thus it is becoming increasingly important to establish security policies.

CONCLUSION

What may be trivial in its self-evidence but profound in its truth is that prevention of computer crime is more effective than treatment. At an organisational level this can be achieved by appropriate legislative controls and better management practices. At a social level, the diffusion of ideas about security as part of the cultural infrastructure could reduce the burden placed on the shoulders of computer security managers. Many large organisations are engaging in awareness campaigns that seek to increase understanding of and sensitivity towards computer security amongst the broad base of their employees, rather than merely concentrate on those with responsibility for computer systems. There are initiatives being developed that target the school age population, seeking to educate youngsters in information security as they learn about computing.

Ultimately the need is to have both a higher level of awareness amongst the workforce generally about the costs and benefits of good security, and a framework of computer law and enforcement and good management practice which will provide the necessary support where the more informal systems of checks and balances fails.

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